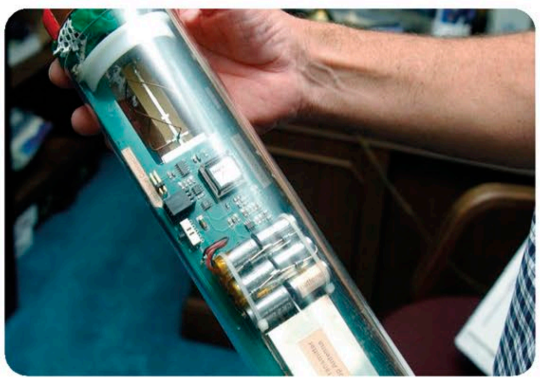
# STUDYING STORMS

## WEATHER PLANES

**Scientists record and analyze data** to forecast the behavior of many weather phenomena, including tropical cyclones (also known as hurricanes). The US National Oceanic and Atmospheric Administration (NOAA) has specially equipped airplanes that fly through these storms to gather data and help scientists understand and predict their activity. Two of NOAA's largest planes, known as Hurricane Hunters, are flying laboratories with radars that can scan the storm and provide real-time information to scientists.

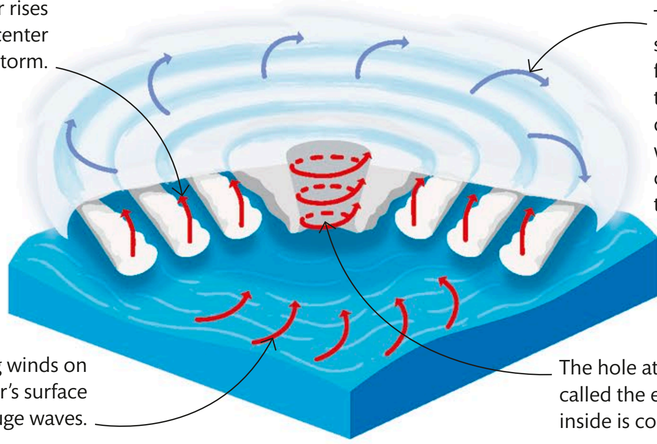
### DROPSONDES

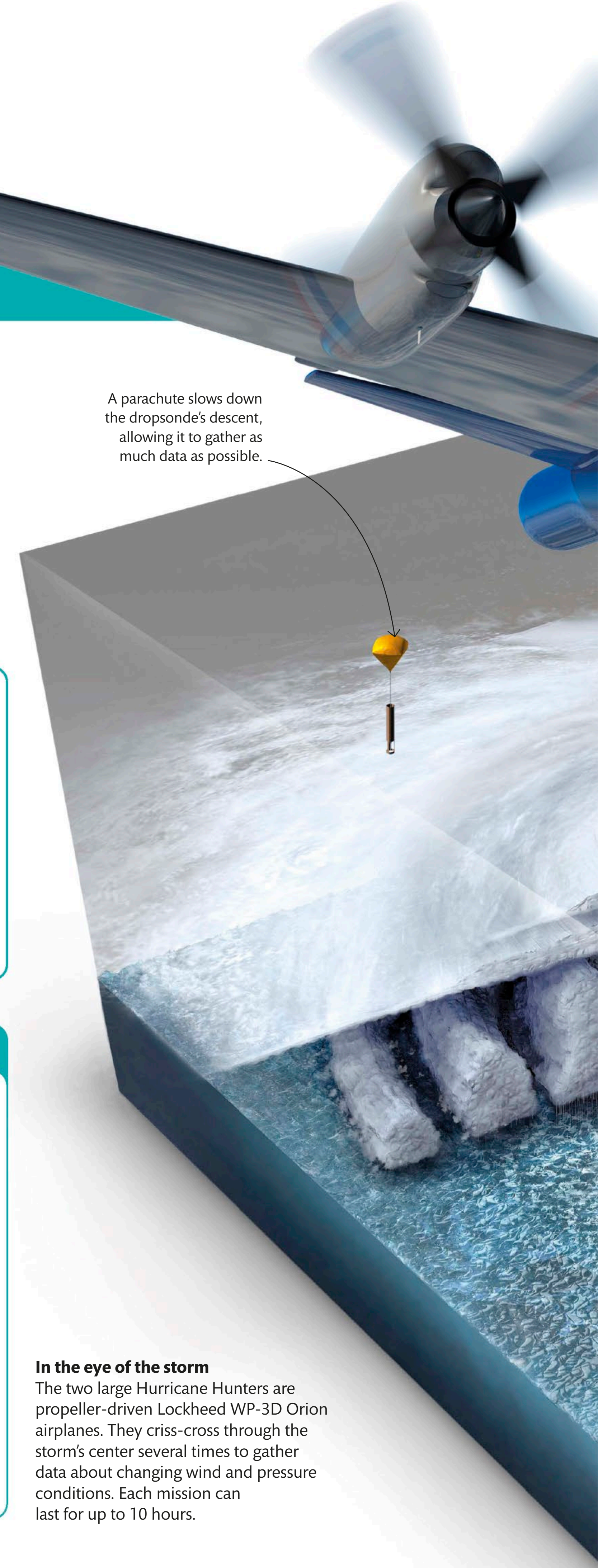
As the Hurricane Hunters fly through the storm, they deploy small capsules called dropsondes. These are packed with sensors that measure pressure, temperature, humidity, and wind direction, and transmit the information back to the plane.



### HURRICANES

Tropical cyclones—also known as hurricanes or typhoons—are large rotating storms of wind, rain, and cloud. They form over tropical oceans where warm, rising air causes an area of low pressure. The air around the center then spirals inward and up, forming bands of rainclouds that are buffeted around in a spiral pattern by ferociously strong winds.




**In the eye of the storm**
The two large Hurricane Hunters are propeller-driven Lockheed WP-3D Orion airplanes. They criss-cross through the storm's center several times to gather data about changing wind and pressure conditions. Each mission can last for up to 10 hours.